UNIVERSITY EXAMINATION · COMPUTE_ENG_EXAM
SUBJECT: ADVANCED ALGORITHMS | SEMESTER 7
Duration: 3 Hours | Max Marks: 80
Instructions: Attempt all questions.

Q1(a) Describe Convolutional Neural Networks for image classification with examples and diagrams. [10 Marks]
Q1(b) Describe Recurrent Neural Networks and Vanishing Gradient with examples and diagrams. [10 Marks]
Q1(c) Describe Long Short Term Memory Networks and with examples and diagrams. [10 Marks]
Q1(d) Describe Autoencoders Denoising and Variational Architectures with examples and diagrams. [10 Marks]
Q1(e) Describe Generative Adversarial Networks Generator Discriminator Loss with examples and diagrams. [10 Marks]
Q2(a) Describe Gradient Descent Optimization Adam Stochastic RMSprop with examples and diagrams. [10 Marks]
Q2(b) Describe Backpropagation Error Signal Chain Rule Derivatives with examples and diagrams. [10 Marks]
Q2(c) Describe Activation Functions ReLU Sigmoid Softmax Tanh with examples and diagrams. [10 Marks]
Q2(d) Describe Regularization Techniques L1 L2 Dropout Weight with examples and diagrams. [10 Marks]
Q2(e) Describe Transfer Learning Fine Tuning Pretrained Backbone with examples and diagrams. [10 Marks]
Q3(a) Describe Transformer Self Attention Multi Head Scaled with examples and diagrams. [10 Marks]
Q3(b) Describe Batch Normalization Layer Normalization Covariate Shift with examples and diagrams. [10 Marks]
Q3(c) Describe Hyperparameter Tuning Learning Rate Schedule Grid with examples and diagrams. [10 Marks]
Q3(d) Describe Loss Functions Cross Entropy Mean Squared with examples and diagrams. [10 Marks]
Q3(e) Describe Object Detection YOLO Faster RCNN Feature with examples and diagrams. [10 Marks]